

Mansakha — Mind Matters, We're Listening

A dual-interface digital platform that gives SC/ST Prevention of Atrocities Act victims one continuous line of support — mental health monitoring, legal aid, financial relief, and protection — while giving the ten-odd government offices around each case a shared, real-time coordination layer they don't have today.

Problem Statement

SIH26094 — Victim support & multi-agency case coordination under the SC/ST (PoA) Act

Category

Software — Government-tech / Social good

Track

Full-stack prototype, working end to end

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01 Proposed Solution

The problem

A victim under the SC/ST (Prevention of Atrocities) Act, 1989 does not deal with one office — they pass through the police station that registers the FIR, the District Welfare Officer who sanctions relief, DLSA for legal aid, the District Collector's Vigilance & Monitoring Committee, and — if the case closes — a rehabilitation centre. Each office keeps its own paper trail. The victim re-tells their story at every desk, has no way to see whether their compensation, lawyer, or protection request is even being looked at, and a genuine threat raised with one office often never reaches the office actually positioned to act on it. Mental health support, when it exists at all, is disconnected from all of it.

The idea

Mansakha is one victim-facing mobile app backed by one shared coordination backend that every relevant government office logs into through its own role-specific web portal. A case, once registered, carries a single continuous record — docket number **DOC-XXXXXX** — that every office reads from and writes to its own corner of, instead of each keeping a separate file. The victim gets one login, one app, and a real-time view of their own case's actual progress; officials get a queue of exactly the work that is theirs, with the proof, the deadline, and the escalation path attached.

Victim side

A continuous, private line of support

Daily emotional check-ins scored by an AI pipeline, a counsellor automatically assigned and reachable by chat, guided wellness exercises, and a one-tap emergency button — inside the same app that tracks their legal case.

Official side

Ten role-specific portals, one shared truth

Ministry, District/State Administration, Data Operator, Counsellor, Investigating Officer, District Welfare Officer, DLSA Coordinator, Protection Officer, District Collector, and Rehabilitation Officer each see only their own queue — every queue is a view into the same case record.

Coordination layer

Referrals, tasks, automatic escalation

A structured referral moves a case from one office to the next. A missed deadline raises itself as a task on the District Collector's desk without anyone having to notice and chase it.

Innovation and uniqueness

01

Proof travels with the case

A victim uploads caste certificate / FIR copy / bank passbook once, against the specific relief type they need. The office that actually verifies that kind of proof in real life reviews it, and accepting it auto-creates the downstream referral. No re-submission, no photocopying the same document for five desks.

02

Two axes of risk, kept separate

An AI-derived Distress Score (is this person okay, mentally) and a rule-based Threat Tier (is this person in physical danger from the accused) are computed independently and never conflated — each drives a different office's response.

03

Escalation is structural, not a hope

Every deadline-bound action is checked by a background worker. If it lapses, a task lands on the District Collector's desk automatically — the Vigilance & Monitoring Committee's real function, made structural instead of relying on memory.

04

Compensation the victim can actually see

The app auto-suggests the applicable statutory relief category and amount (grounded in the Dr. Ambedkar National Relief Scheme's real tiers) and shows a live 3-stage payment tracker tied to the case's own real progress.

02 Technologies to be Used

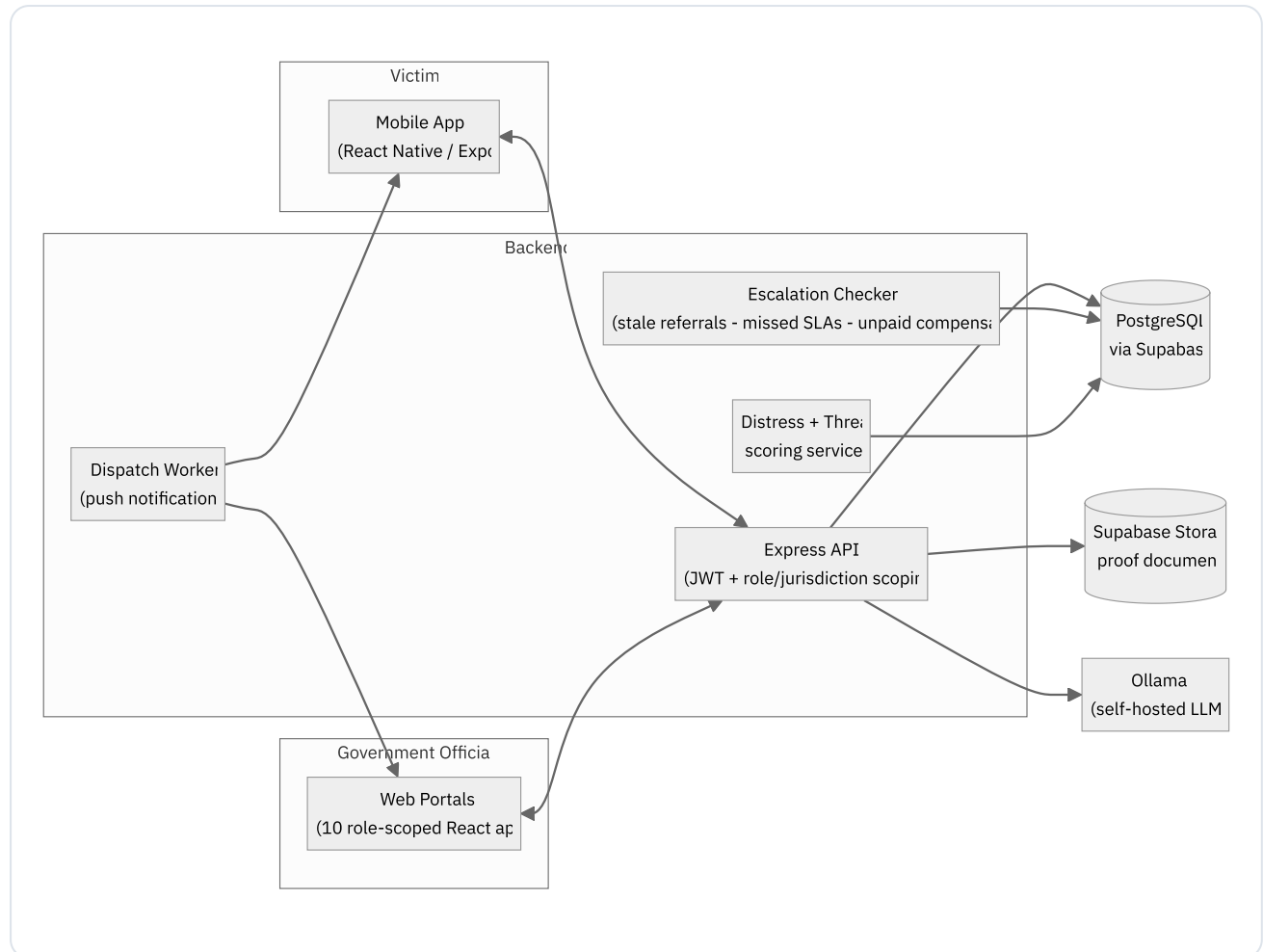
A conventional, provable stack was chosen deliberately over a novel one — this is government-facing infrastructure carrying sensitive case data; every layer below is boring on purpose, and the full system already runs end to end.

LAYER	CHOICE	WHY
Victim mobile app	React Native Expo React Query	One codebase for Android, iOS, and a web build — reaches a low-connectivity, low-end-device population without a separate native team per platform.
Official web portals	React Vite Tailwind CSS	Ten independent role-scoped portals sharing one design system, fast to extend without touching the others (each keeps its own hooks/auth, deliberately not shared).
Backend API	Node.js Express JWT auth	Role- and jurisdiction-scoped REST API; every request re-checks the caller's live role grants — a reassignment takes effect immediately, no re-login required.
Database	PostgreSQL Supabase Supabase Storage	Relational integrity for a system where a case has one true record touched by many roles; Storage holds proof documents behind signed, time-limited URLs, never public.
AI / distress pipeline	Ollama (local LLM) Custom scoring service	Sentiment, voice-stress, emotion and engagement signals combine into a Low→Critical risk level; runs against a self-hosted model so no victim's check-in data leaves the deployment's own infrastructure.
Notifications	Expo Push Background dispatch worker	A Critical alert, an SOS, or a missed SLA reaches the right official's phone without them having to be watching a dashboard.
Location (best-effort)	expo-location	Captured only on an explicit action (a threat report, an emergency call) and only with permission — never tracked passively.

03 Methodology & Process for Implementation

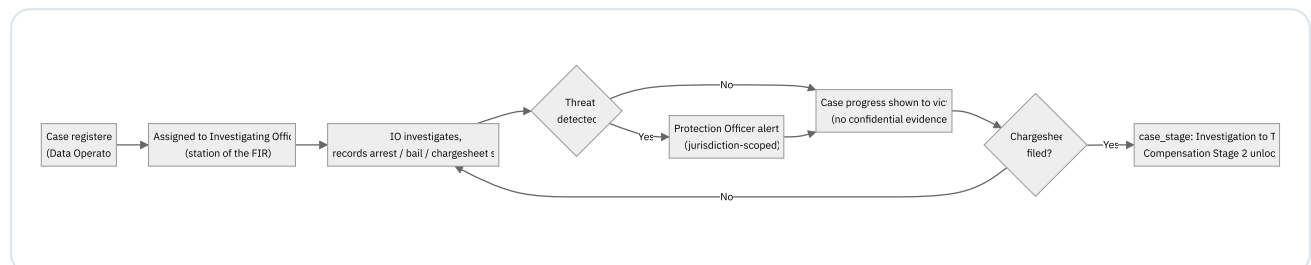
The system is built as one shared case record surrounded by role-scoped queues. The diagrams below are the actual working architecture and workflows, not a target design.

System architecture

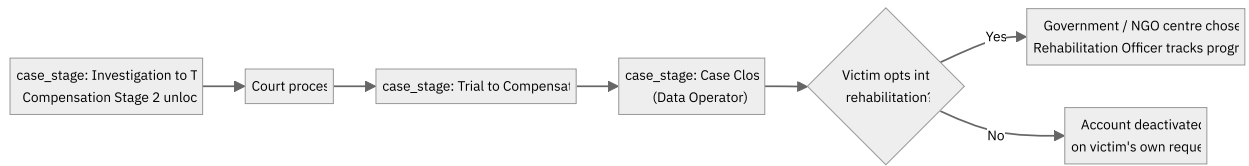


Every role's portal and the victim app talk to one API, which is the only thing that touches the database — no portal reads another portal's data directly.

A. Case lifecycle — registration to closure

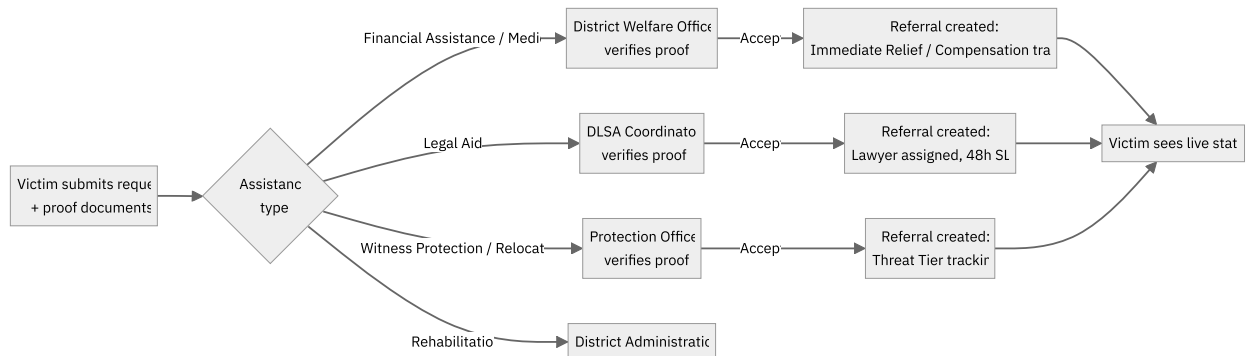


Every open case is assigned to the station that filed its FIR — this is IO's own queue, not a District-Admin-created referral like the roles below.



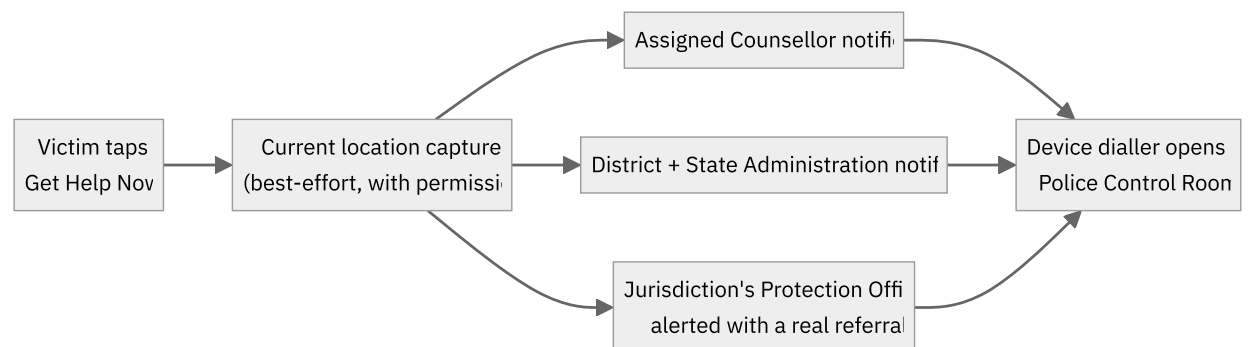
case_stage is the one shared fact every role's own screen reads — DWO's compensation stages, IO's investigation record, and the victim's own Case Details all move together.

B. Request Assistance — proof reaches the right verifier



The specialist who actually verifies that kind of proof in reality is the one who reviews it in the app — not a single generalist office deciding on everything.

C. Emergency / SOS



One tap fans out to every office that can actually help, in parallel, before the victim ever has to explain the situation to any of them.

Roles directory — every login, why it exists, what it does

Each role below maps to a real office or statutory function under the Act; none is invented for the sake of having more logins.

Ministry

National

National nodal authority under the Act.

DOES

Creates every official account, sets system-wide configuration (case types, languages, intervention proof requirements), unrestricted read access for oversight.

WHY IT EXISTS

Someone has to provision every other login and own data-driven configuration without redeploying code.

District / State Administration

Oversight

District/State-level administrative machinery under the Act.

DOES

Case intake oversight, manual cross-agency referral, full read visibility on Request Assistance (decides only the Rehabilitation type itself now).

WHY IT EXISTS

Someone needs district-wide visibility across every role's queue, even after most decision authority moved to the specialist office that actually verifies each kind of proof.

Data Operator

Intake

Front-desk / helpline intake staff.

DOES

Registers a new case, issues the victim's docket ID and credentials, is the only role that can mark a case Closed.

WHY IT EXISTS

Case closure is deliberately a single, auditable action distinct from every other case_stage transition.

Counsellor

Mental health

Assigned mental-health support professional.

DOES

Chats with assigned victims, receives Critical distress-score alerts and SOS notifications, reviews AI-drafted check-in notes.

WHY IT EXISTS

A victim's psychological state is monitored continuously, not only when they think to ask for help.

Investigating Officer

Reinstated

The police officer who registered the FIR, at that specific station.

DOES

Reviews the assigned case, investigates and collects evidence internally, records victim-safe progress (accused status, chargesheet status), alerts Protection Officer on a detected threat, marks investigation complete - advancing the case to Trial.

WHY IT EXISTS

The one role with real statutory custody over arrest/bail/chargesheet facts; scoped to a real police station, not just a district.

District Welfare Officer

Relief

District Social Welfare Department.

DOES

Approves fast Immediate Relief (money/essential support) and verifies + tracks the larger statutory Compensation in 3 payment stages tied to real case progress.

WHY IT EXISTS

The PoA Act's own relief and compensation scheme names this office as the disbursing authority.

DLSA Coordinator

Legal aid

District Legal Services Authority.

DOES

Assigns panel counsel within a 48-hour SLA, receives victim feedback on assigned lawyers.

WHY IT EXISTS

Free legal aid for an SC/ST victim is a NALSA entitlement gated on caste proof, not income - DLSA is who actually administers it.

Protection Officer

Safety

Witness/victim protection function under the Act.

DOES

Jurisdiction-scoped threat-tier tracking, receives IO's threat alerts and the victim's own emergency SOS with location.

WHY IT EXISTS

Physical safety is judged and acted on separately from a victim's psychological distress score.

District Collector

Escalation

Chair of the Act's real Vigilance & Monitoring Committee.

DOES

Receives every automatically-escalated task - a stale referral, a missed SLA, an unpaid compensation stage - and issues directives.

WHY IT EXISTS

Matches the Committee's real statutory oversight function almost exactly.

Rehabilitation Officer

Post-closure

Staff at a Government (incl. DMHP) or NGO rehabilitation centre.

DOES

Scoped to one specific centre, tracks a victim's ongoing rehabilitation progress after case closure, can mark a victim as having left the centre.

WHY IT EXISTS

Support does not end at a verdict - a victim who opts in gets a genuinely new, centre-specific dashboard experience.

04 Potential Impact on the Target Audience

The target audience is two-sided: the victim who today re-tells their story at every desk, and the official who today works from a paper trail with no visibility into what any other office has done.

1

login, one continuous case record, instead of a separate file at every office

10

coordinated official roles working the same case instead of in isolation

48h

SLA on legal-aid counsel assignment, enforced by automatic escalation

3

statutory compensation stages the victim can actually see, tied to real case progress

Social benefit

- A victim is never asked to re-narrate the incident to a new desk — the case record already carries what the previous office recorded.
- Continuous mental-health monitoring reaches victims who would not otherwise think to ask for counselling.
- A threat raised in one office reaches the office that can act on it without depending on an individual officer remembering to make a call.

Economic & administrative benefit

- Compensation staged against real, verifiable case milestones reduces both premature disbursement and indefinite pendency.
- Structural escalation replaces manual case-tracking registers District Collector offices otherwise maintain by hand.
- Digitised proof documents remove repeated physical photocopying and courier movement between offices.

For officials

- Each portal shows exactly the queue that is that role's own responsibility.
- A reassignment (a new jurisdiction, a new rehabilitation centre) takes effect on the very next request, no manual handover required.

Scalability

- Every role is jurisdiction- or station-scoped by data, not by a separate deployment.
- The same architecture generalises to any Act with a comparable multi-agency victim-support structure.

05 Feasibility and Viability

The system described above is not a concept — it is a working prototype today, covering the victim app, all ten role portals, the shared referral/task/escalation backend, and the AI scoring pipeline. The table below is an honest accounting of what remains between this prototype and a real deployment.

CHALLENGE	RISK	MITIGATION STRATEGY
Interoperability with existing government systems (NHAA Helpline, Integrated Portal, eCourts)	Medium	Docket numbers are already designed to be reused from NHAA/Integrated Portal where available; a dedicated ingestion adapter is the next integration surface rather than a redesign.
Adoption by officials used to paper registers	Medium	Each role's portal is scoped to a small, single-purpose queue, specifically to keep the learning curve short for one office at a time.
Sensitive data handling (proof documents, case background, distress scores)	High	PII lives in a table separate from the scoring/alert pipeline; proof documents are served only via short-lived signed URLs, never public; every read/write is audit-logged.
Connectivity in rural/low-bandwidth areas	Medium	The victim app is built mobile-first on Expo/React Native, with SMS-based check-in prompts as a deliberately lighter-weight fallback channel already in the design.
Self-hosted AI inference cost and latency at scale	Low	The one time-critical action (emergency SOS) is deliberately built to bypass the AI pipeline entirely - distress scoring runs asynchronously and never blocks a safety-critical response.
Legal/administrative authority to onboard real government officials	High	Requires a sponsoring department (State Social Welfare / Home Department) to mandate onboarding - a policy dependency outside the software itself, common to any government-facing platform.

The prototype's core bet is narrow and testable: that giving each existing office a smaller, sharper queue — instead of asking any single office to do more — is what actually gets a stalled multi-agency case moving. Every workflow diagrammed above already runs against a real database with real role-scoping, not a mockup.